



**USING ARTIFICIAL INTELLIGENCE TO BETTER UNDERSTAND NATURAL
INTELLIGENCE: DEVELOPING A SYNTHETIC EEG TESTBED FOR EVALUATING
BRAIN STATE DETECTION ALGORITHMS**

A Senior Honors Thesis

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Preface

This paper represents the culmination of my senior honors research project in computational neuroscience. Over the course of this semester, I explored the intersection of signal processing, applied mathematics, machine learning, and neuroinformatics to develop a robust testbed for evaluating algorithms designed to detect brain activity patterns in EEG data. My primary focus was on generating, processing, and interpreting synthetic EEG data to create a controlled environment for testing detection methodologies across varying noise conditions.

By applying mathematical models and signal processing techniques, I aimed to build a framework for evaluating the robustness of neural oscillation detection algorithms through an iterative process of hypothesis testing, validation, and refinement. This research has deepened my technical expertise and provided an opportunity to contribute meaningfully to the growing field of computational neuroscience. I am especially grateful to my mentor, Daniel Scott, whose guidance and encouragement were invaluable throughout.

Abstract

Neural oscillations, characterized by distinct frequency bands — delta (0.5–4 Hz), theta (4–8 Hz), alpha (8–12 Hz), beta (12–30 Hz), and gamma (30–100 Hz) — are integral to brain function. Computational methods for analyzing these oscillations can provide profound insights into brain dynamics and their relationship to cognitive and physiological processes. This research focuses on the development of a synthetic EEG testbed to evaluate the performance of brain state detection algorithms under various noise conditions.

Using a three-pronged approach to noise simulation (white noise, pink noise, and real EEG background), we generated synthetic datasets with known ground truth signals that mimic real neural oscillations. We then employed signal processing techniques, including Short-Time Fourier Transform (STFT) and Discrete Wavelet Transform (DWT), to evaluate how accurately these frequency bands could be detected across different noise types and levels. Receiver Operating Characteristic (ROC) analysis was used to systematically assess detection sensitivity and specificity and to identify optimal thresholds for each frequency band under varying noise conditions.

The results demonstrate that neural oscillations exhibit varying degrees of resilience to noise: alpha waves showed the highest detection robustness, while gamma oscillations proved most vulnerable under broadband noise conditions. This testbed provides a foundation for evaluating more sophisticated detection and prediction algorithms, with potential applications in clinical diagnostics, cognitive research, and neural engineering.

Introduction

Electroencephalography (EEG) is a non-invasive technique for measuring electrical activity of the brain via electrodes placed on the scalp (Craik et al., 2019). These signals reflect the synchronized activity of large neuronal populations and manifest as oscillations across distinct frequency bands, each associated with different cognitive and physiological states: delta (0.5–4 Hz) with deep sleep, theta (4–8 Hz) with memory formation and drowsiness, alpha (8–12 Hz) with relaxed wakefulness, beta (12–30 Hz) with active thinking and focus, and gamma (30–100 Hz) with higher-order cognitive processing and perceptual integration (Bashivan et al., 2015).

Time-frequency decomposition allows researchers to analyze these bands separately, yielding insights into brain function under different conditions. Numerous studies have established correlations between power fluctuations in specific frequency bands and cognitive states. Increased alpha power, for example, is associated with relaxation and inattention, while beta power typically rises during tasks requiring sustained concentration (Roy et al., 2019).

Accurately detecting and predicting transitions between brain states has significant implications across several domains (Zhang & He, 2018), including:

1. Clinical diagnostics for neurological disorders such as epilepsy, where abnormal oscillatory patterns may precede seizures
2. Brain-computer interfaces (BCIs) that rely on identifying specific brain states for device control

3. Neurofeedback therapy, where real-time brain state detection enables patients to modulate their own neural activity
4. Sleep research, where distinct oscillatory patterns characterize different sleep stages

Despite the importance of accurate brain state detection, evaluating algorithm performance remains challenging. Real EEG data is inherently noisy, with signals of interest often obscured by artifacts and background activity. Crucially, the ground truth regarding when specific brain states occur is frequently uncertain, making objective accuracy assessment difficult.

This highlights the need for a controlled testing environment — a synthetic EEG testbed — where algorithms can be evaluated against known ground truth under varying noise conditions. Such a testbed allows researchers to systematically compare detection algorithms before applying them to real-world data. Signal processing techniques like Fourier and wavelet transforms enable decomposition of EEG signals into constituent frequencies, while machine learning models allow for pattern recognition within these datasets. Both are central to the methods employed in this thesis.

The goal of this work is to develop and validate a synthetic EEG testbed that simulates realistic neural oscillations embedded in multiple types of noise. By controlling the signal-to-noise ratio across white noise, pink noise, and real EEG background conditions, this testbed enables objective evaluation of detection algorithm performance through ROC analysis.

This work lays the foundation for future development of more sophisticated algorithms, including predictive models that can anticipate upcoming brain states from preceding signals.

Methodology

Synthetic EEG Signal Generation:

The foundation of this research is the generation of synthetic EEG data that mimics real neural oscillations while incorporating realistic noise. Using computational models, we simulated delta, theta, alpha, beta, and gamma waves with characteristic frequencies and amplitudes, and introduced Gaussian noise to add variability and replicate real-world recording conditions.

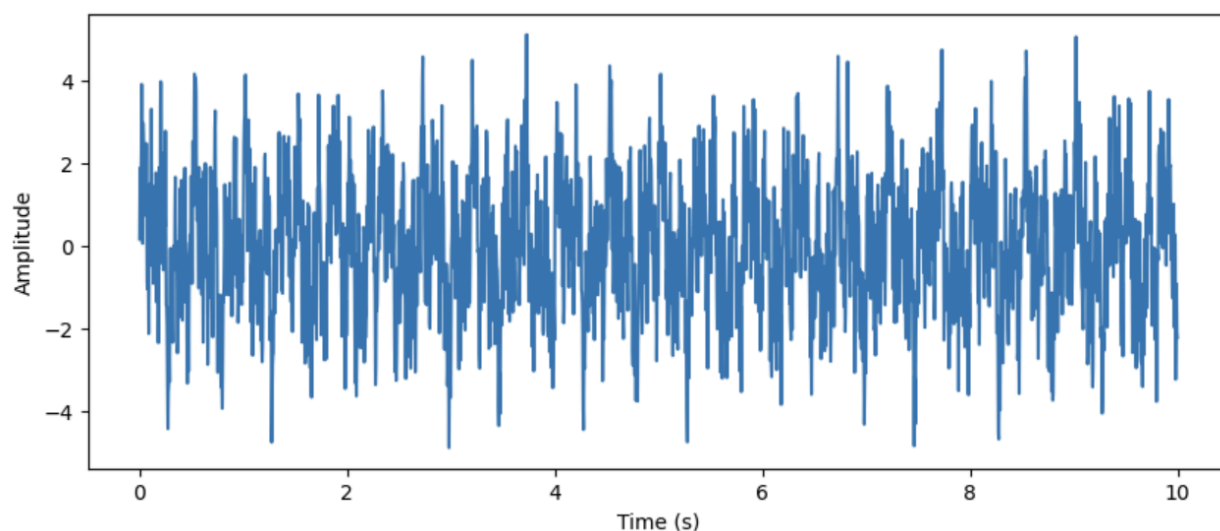


Figure 1. Noisy Synthetic EEG Data

EEG = electroencephalogram

The synthetic signal shown in Figure 1 was generated by superimposing five sinusoidal waves representing the standard EEG frequency bands, all present simultaneously throughout a 10-second recording (2500 samples at 250 Hz). Specifically, we used 2.25 Hz for delta, 6 Hz for theta, 10 Hz for alpha, 21 Hz for beta, and 65 Hz for gamma — representative center frequencies

within each band's range. All five components were overlaid at equal amplitudes to form a composite signal. Gaussian noise (mean = 0, standard deviation = 1) was then added at a scaling factor of 0.5 to simulate realistic recording conditions. Ground truth for each frequency band was stored as a binary matrix indicating presence (1) or absence (0) of each oscillation at each time point, with interpolated values during transition periods.

To evaluate detection performance under varying conditions, we embedded the synthetic signals in three types of noise:

- White Noise: Equal power across all frequencies, generated using a Gaussian distribution.
- Pink Noise: Power concentrated in lower frequencies following a $1/f$ distribution, more closely resembling background EEG activity.
- Real EEG Background: Circular shifting of actual EEG recordings to create realistic, structured noise.

Pink Noise Generation & Analysis:

Unlike white noise, which has a uniform spectral distribution, pink noise better simulates realistic EEG background activity due to its higher power at lower frequencies. We implemented a circular shifting approach to generate diverse but reproducible noise patterns across iterations.

The pink noise generation formula is:

The pink noise generation formula is defined as: $x_{noise}(t, s) = \sum_{f \in F} \frac{A_f}{\sqrt{f}} \sin(2\pi f t + \phi_f) \cup s$

where $F = \{0.5, 1, 2, 4, 8, 12, 16, 20, 30\}$ Hz represents frequencies commonly observed in EEG, A_f is the amplitude following a $1/f$ distribution consistent with real EEG power spectral density, ϕ_f is a fixed phase term for each frequency, and $\oslash_s$ denotes circular shifting by s steps (ranging from 0 to 83,325 data points).

For each simulation, 100 noise patterns were generated by sampling shift values across the full range, ensuring reproducibility of individual patterns, sufficient diversity to represent real-world variability, and non-repeating noise structure across conditions. Progressive noise levels from 0.1 to 100.0 were applied to assess algorithmic robustness under increasingly challenging conditions.

Frequency Band Discrimination:

To differentiate between frequency bands in the noisy synthetic data, we implemented a frequency masking approach. By defining the boundaries of each band, we created masks to filter and reconstruct the corresponding frequency components. This allowed visual verification that signal generation was functioning as intended. The approach was further validated by comparing reconstructed signals against the original synthetic data using the Discrete Wavelet Transform (DWT).

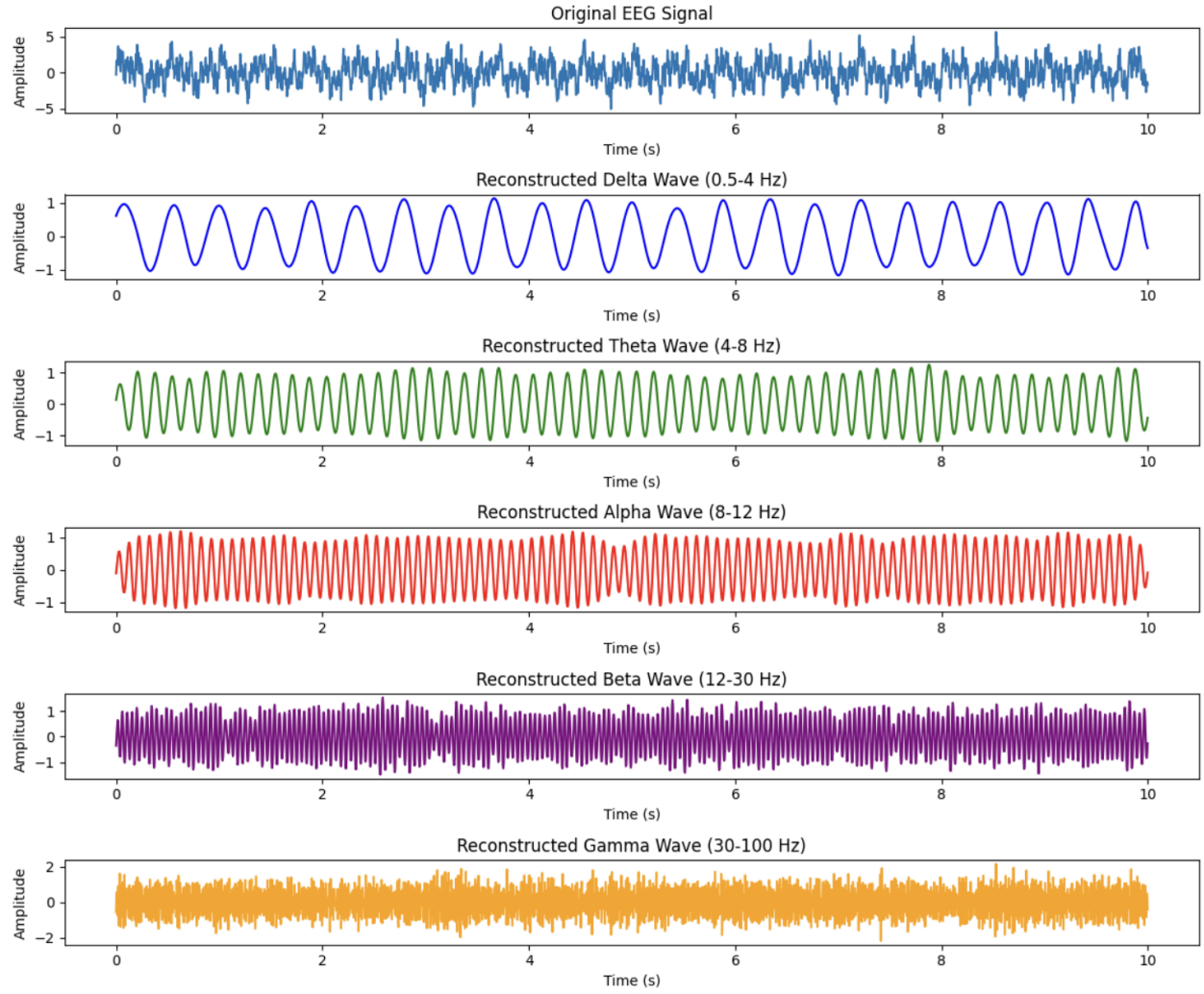


Figure 2. Frequency Band Discrimination

Wavelet Decomposition:

Single-level DWT decomposition was performed using the Daubechies wavelet to further validate the frequency band discrimination. By reconstructing the noisy synthetic EEG into continuous signals, we confirmed that the DWT could effectively capture the temporal structure of neural oscillations.

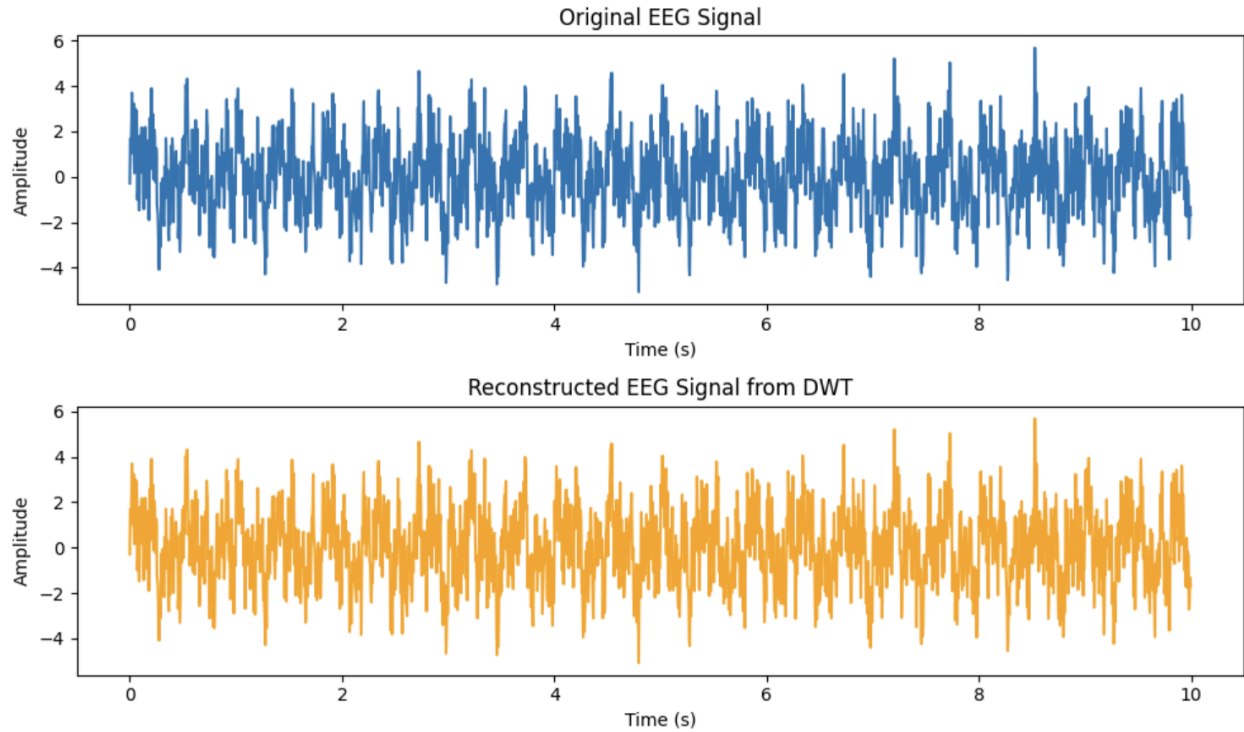


Figure 3. Reconstructed EEG Signal

Short-Time Fourier Transform (STFT):

The STFT was applied to the synthetic EEG data to visualize frequency content over time. The resulting spectrogram representation revealed the temporal evolution of different frequency bands. By concatenating frequency bands into a continuous signal, we assessed how well the STFT could detect and separate bands in the time-frequency domain.

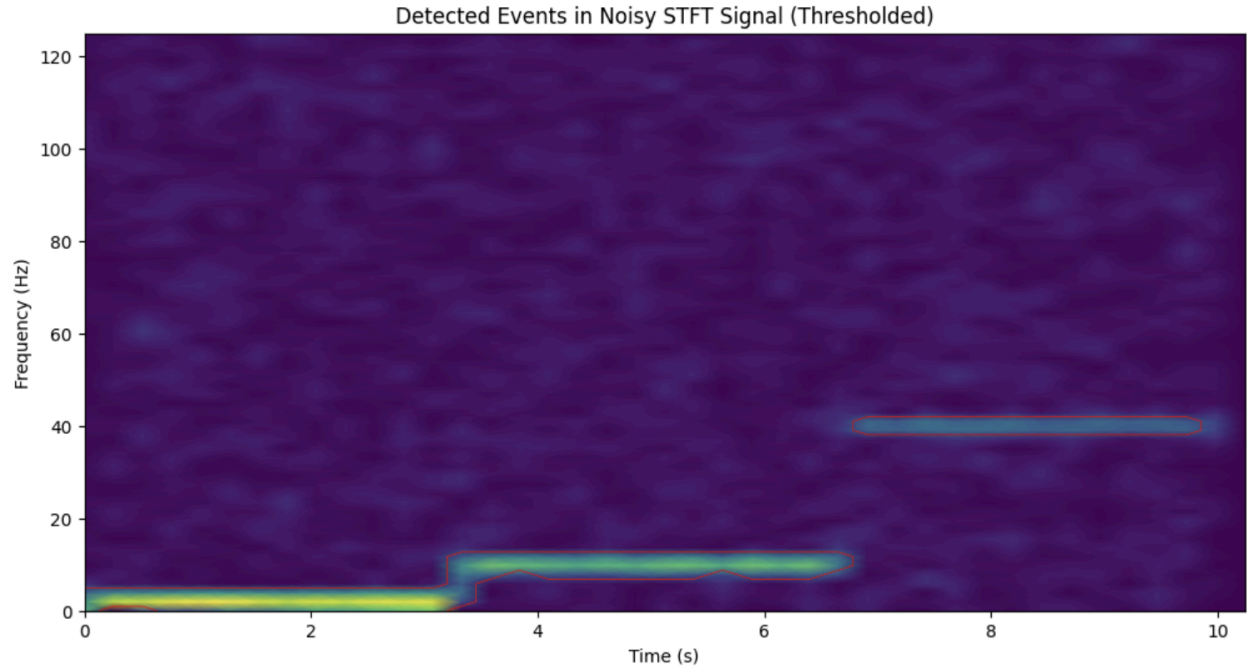


Figure 4. STFT Detected Events in Noisy STFT Signal

Time-Series Chunking for EEG Data:

To analyze neural oscillations effectively, we segmented the EEG data into fixed-duration chunks (3.333 seconds at 250 Hz). Gaussian envelopes were applied at chunk boundaries to smooth transitions and eliminate edge artifacts, enabling continuous signal analysis while preserving the frequency characteristics needed for wavelet detection.

Frequency Band Transition Detection:

Each frequency band was independently extracted using STFT with a Hanning window of 250 samples (1 second) and 90% overlap between adjacent windows, enabling extraction of band power at each time point. Band-specific smoothing parameters were tailored to each frequency range; gamma oscillations, for example, received additional Gaussian envelope smoothing to compensate for their elevated susceptibility to noise.

ROC-Based Frequency Band Detection:

To quantitatively evaluate detection performance, we developed a comprehensive ROC analysis pipeline enabling systematic assessment of sensitivity and specificity across noise conditions and frequency bands.

For each band, ground truth labels were established by tracking the exact temporal occurrence of each frequency component in the synthetic data. A threshold-based detection approach was then applied, classifying time points where band power exceeded a given threshold as positive detections. For each band and noise level, 20 threshold values were tested, ranging from zero to three times the noise level — an adaptive range designed to ensure appropriate sensitivity across widely varying signal-to-noise conditions.

For each threshold, the true positive rate (sensitivity) and false positive rate ($1 - \text{specificity}$) were computed by comparing detector output against the ground truth labels. Precision metrics were also calculated to identify thresholds that maximized both sensitivity and specificity simultaneously.

Results

ROC curves provided a visual representation of detection performance, while the area under the curve (AUC) served as a single summary metric for comparison across conditions. AUC values near 1.0 indicate excellent detection performance; values near 0.5 indicate chance-level detection. By tracking AUC across noise levels, we identified the specific thresholds at which detection became unreliable for each frequency band.

The ROC analysis revealed substantial differences in noise resilience across frequency bands and noise types. Results across 100 noise patterns enabled identification of band-specific optimal thresholds, providing a foundation for adaptive detection algorithms that could modulate sensitivity based on signal quality and target frequency.

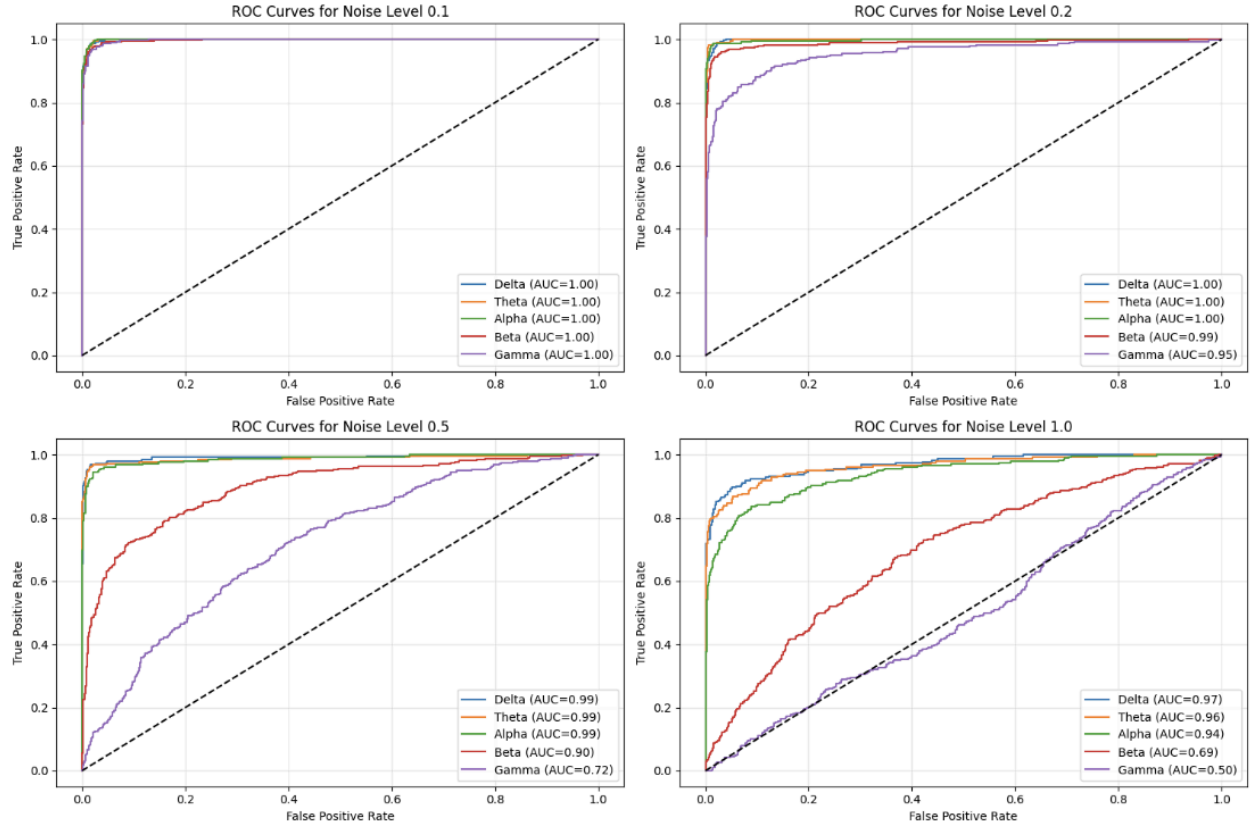


Figure 5. ROC Analysis of Synthetic EEG Data with White Noise at Various Noise Levels

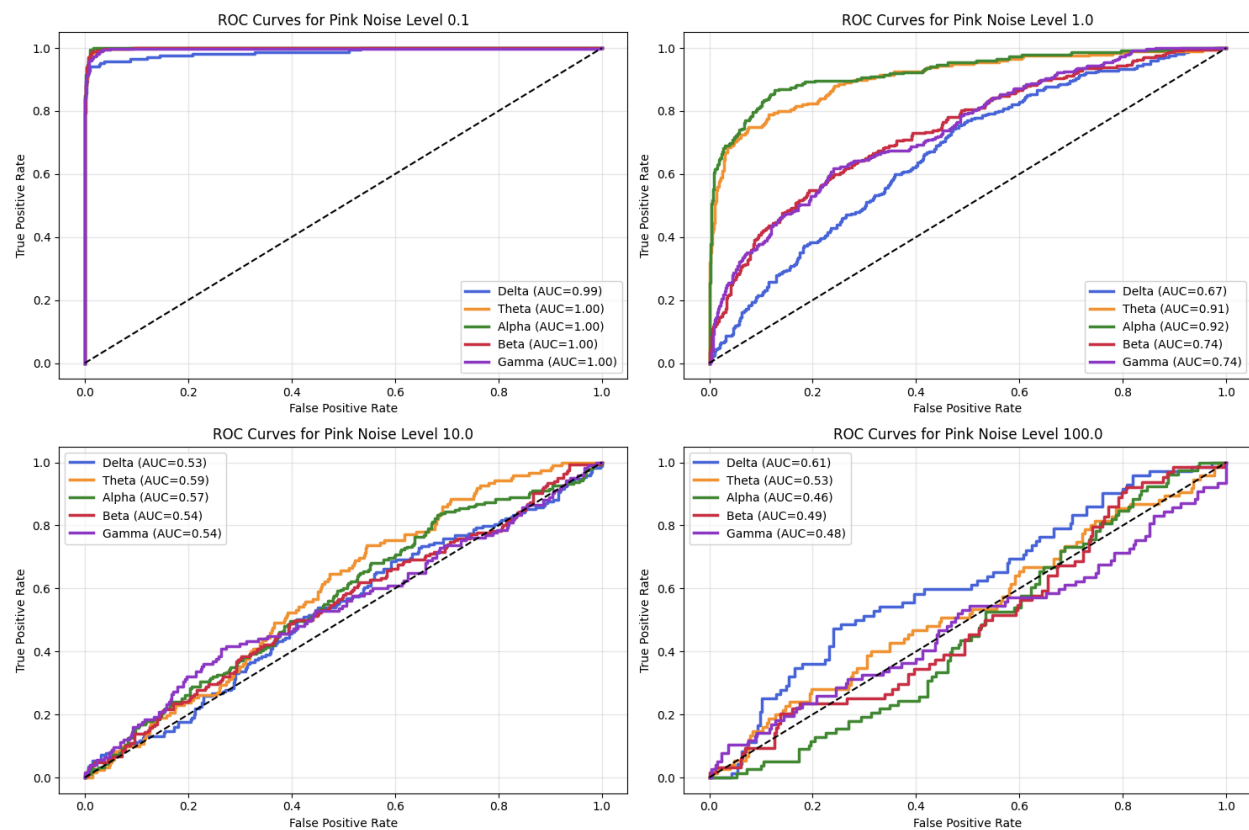


Figure 6. ROC Analysis of Synthetic EEG Data with Pink Noise at Various Noise Levels

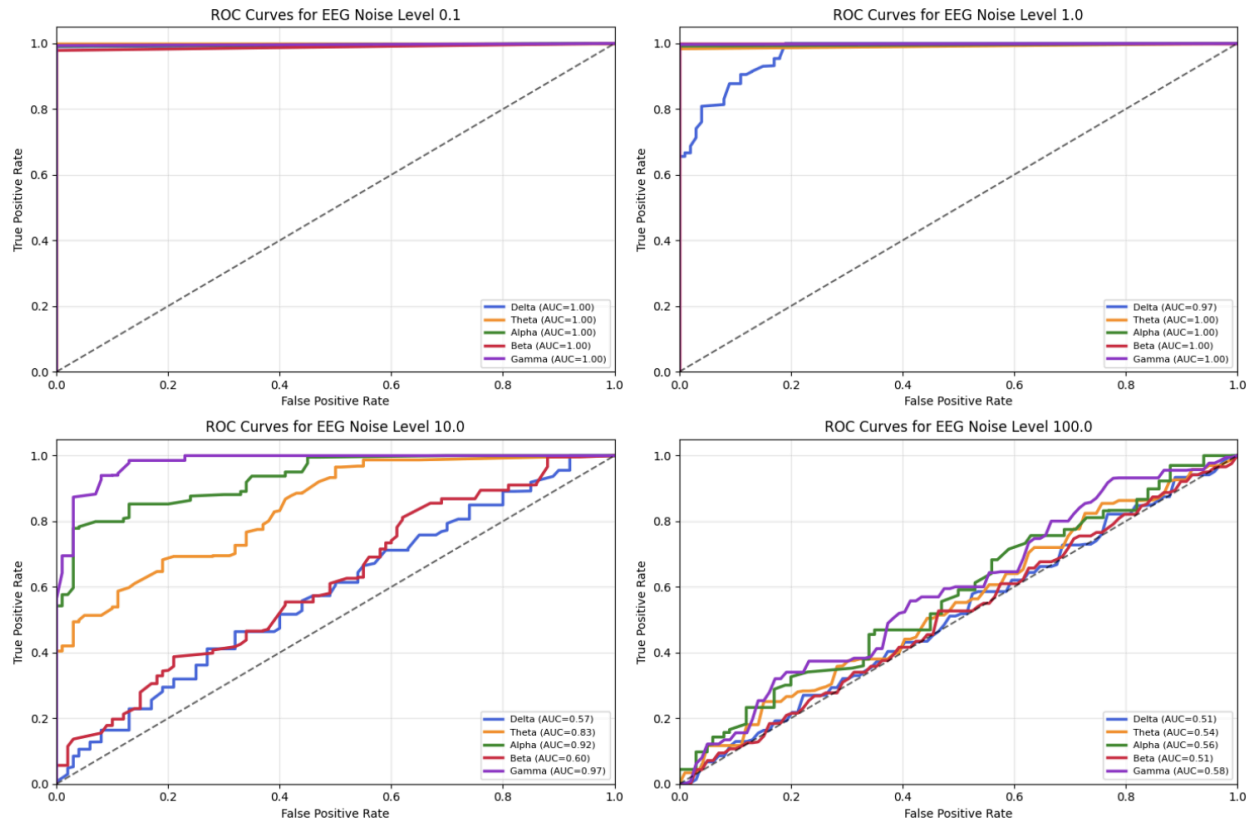


Figure 7. ROC Analysis of Synthetic EEG Data with Real EEG Data as Noise at Various Noise Levels

White Noise Conditions (Figure 5):

At noise level 0.1, all five bands achieved near-perfect detection ($AUC = 1.00$). As noise increased, gamma was the first to degrade: by noise level 0.2 it had dropped to $AUC = 0.95$, and by noise level 1.0 it had reached chance-level performance ($AUC = 0.50$). Beta showed moderate degradation across the same range ($AUC = 0.99 \rightarrow 0.69$), while delta, theta, and alpha remained highly resilient, maintaining $AUC \geq 0.94$ even at noise level 1.0. The rapid deterioration of gamma under broadband white noise suggests that high-frequency oscillations are particularly sensitive to spectrally uniform interference.

Pink Noise Conditions (Figure 6):

At noise level 0.1, all bands again achieved excellent detection ($AUC \geq 0.99$). By noise level 1.0, however, the pattern reversed relative to white noise: delta became the most vulnerable band ($AUC = 0.67$), while alpha and theta showed the greatest resilience ($AUC = 0.92$ and 0.91 , respectively). This inversion is consistent with the spectral properties of pink noise, which concentrates power in lower frequencies and therefore interferes most directly with delta band detection. By noise level 10.0, most bands had degraded to near chance-level performance, and by noise level 100.0, all bands were effectively at chance ($AUC = 0.46\text{--}0.61$).

Real EEG Background Noise (Figure 7):

Real EEG background noise produced the most distinctive results. Detection remained near-perfect for all bands at noise level 1.0 ($AUC = 0.97\text{--}1.00$), and even at noise level 10.0, gamma showed remarkable resilience ($AUC = 0.97$) while alpha remained strong ($AUC = 0.92$).

Beta and delta degraded more substantially at this level ($AUC = 0.60$ and 0.57 , respectively). By noise level 100.0, all bands approached chance-level detection ($AUC = 0.51$ – 0.58).

The exceptional resilience of gamma under real EEG noise — in sharp contrast to its vulnerability under white noise — is the most noteworthy cross-condition finding. We attribute this to spectral complementarity: real EEG background noise, unlike white noise, has a non-uniform spectral structure that concentrates power in specific bands. At equivalent overall power levels, this means real EEG noise interferes less with gamma-band signals due to minimal spectral overlap in that frequency range. This interpretation is supported by the aggregate AUC comparison at noise level 1.0: white noise yielded a summed AUC of 4.06 across all bands, pink noise 3.98, and real EEG background 4.97.

This finding also points to an important methodological consideration: SNR normalization based on total signal power may not adequately capture the effective interference a given noise type exerts on a specific frequency band. Future evaluations should account for the spectral relationship between signal and noise, not just overall power ratios.

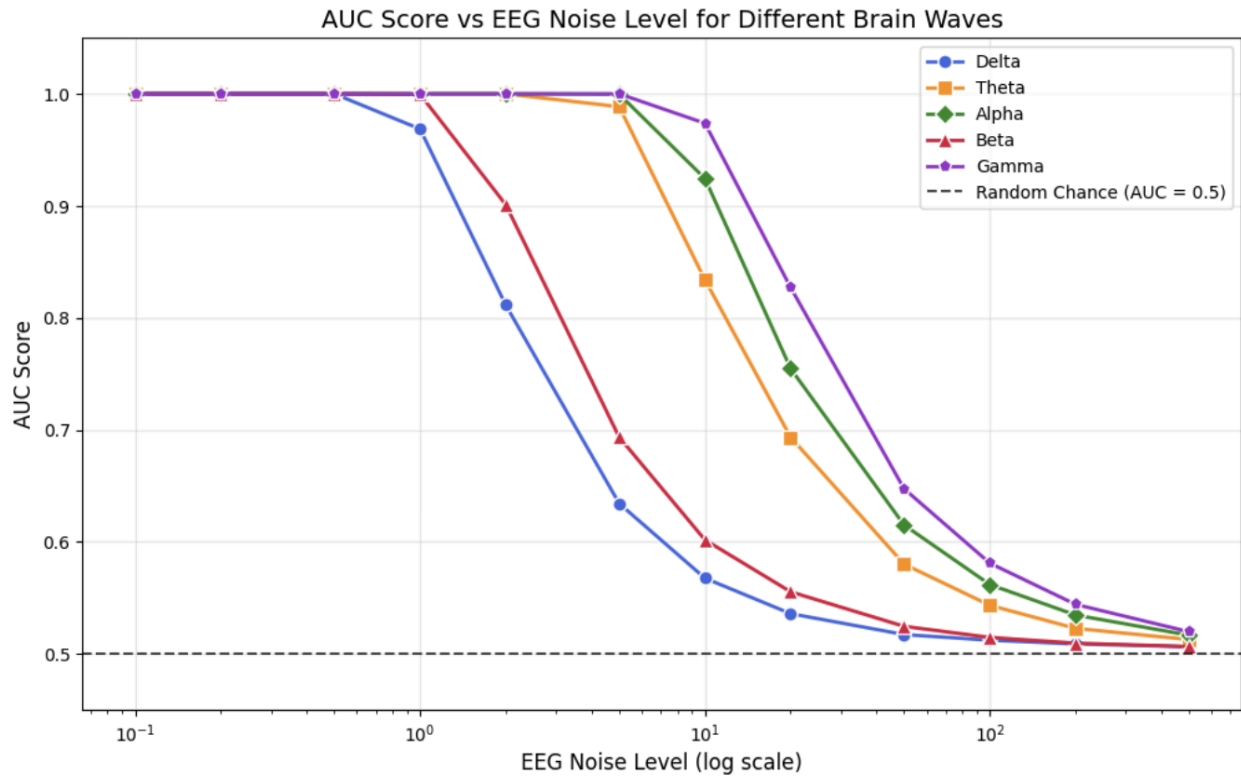


Figure 8. Logistic Graph of AUC Score vs. Pink Noise Level for All Brain Waves

Across all three noise conditions, the relationship between AUC and noise level followed a sigmoidal decay pattern, with band- and noise-type-specific inflection points at which detection performance degraded rapidly. These thresholds provide practical guidance for calibrating detection algorithms to specific application contexts.

Discussion

The results demonstrate that the synthetic EEG testbed successfully provides a controlled environment for evaluating brain state detection algorithms. By embedding known ground truth signals in different types of noise, we established a framework for objectively assessing detection performance — addressing a meaningful gap in the field, where evaluation has historically relied on subjective judgment or uncertain ground truth from real recordings.

The differential noise resilience of frequency bands has direct implications for applied EEG analysis. Alpha oscillations proved most robust across noise types, suggesting they may serve as reliable markers for brain state identification even in noisy environments. Gamma oscillations showed high vulnerability to broadband noise, underscoring the challenges of detecting high-frequency signals in clinical contexts — though their unexpected resilience under real EEG noise suggests this vulnerability is noise-type-dependent rather than absolute.

The primary limitation of the current approach is the simplicity of the detection algorithm: band power thresholding validates the testbed's functionality but does not reflect the complexity of algorithms used in practice. Future work should evaluate more sophisticated methods, including machine learning classifiers and deep learning approaches, within this framework. A second limitation is that the synthetic oscillations are stationary and sinusoidal, and therefore do not capture real neural signal complexity, including cross-frequency coupling, non-sinusoidal waveforms, and non-stationarity. Future iterations should incorporate these features to improve ecological validity.

Conclusion

This research establishes a synthetic EEG testbed for evaluating brain state detection algorithms under varying noise conditions. By generating controlled signals with known ground truth and embedding them in three distinct noise types, we created a framework for objectively assessing algorithm performance through ROC analysis.

The results show that neural oscillations vary considerably in their noise resilience, with alpha waves proving most robust and bands such as delta and gamma exhibiting noise-type-dependent vulnerability. These findings offer practical guidance for researchers and clinicians working with EEG in noisy environments and highlight the importance of matching noise models to the intended application when benchmarking detection algorithms.

This testbed lays the groundwork for several directions in future research: evaluation of deep learning-based detection algorithms that may better handle real-world noise and artifacts; development of predictive models capable of anticipating upcoming brain states from preceding signals; and extension to multi-channel EEG simulations incorporating spatial patterns and source localization challenges. As computational methods continue to advance, standardized evaluation frameworks of this kind will play an increasingly important role in ensuring the reliability and clinical applicability of brain state detection systems.

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